



STUDENT NAME:

TEACHER NAME: BERD CVEK KERJ MAYR MCLS SCHP XIAR
(please circle)

MATHEMATICAL METHODS UNIT 4

SAC 3 (Tech-Active)

Schmidipus the King



Reading Time: 15 minutes

Writing time: 120 minutes

QUESTION AND ANSWER BOOK

<i>Number of Questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
8	8	72

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners, rulers, a protractor, set squares, aids for curve sketching, one bound reference, one laptop with Mathematica and any number of Mathematica files.
- Students are NOT permitted to bring into the examination: blank sheets of paper and/or correction fluid/tape.

Materials supplied

- Question and answer book of 22 pages.

Instructions

- Write your name in the space provided above on this page.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Note: Sections between these symbols are story only and are not needed to answer the questions



A plague has stricken JMSS. The citizens gather outside the palace of their king, Schmidipus, asking him to take action. Schmidipus replies that he already sent his colleague, Kreemond, on a journey to Corkie, the oracle of Education, to learn how to help JMSS. There is little that Corkie does not know! Corkie is a very busy oracle and Kreemond knows that it may take some time to organise a meeting once he arrives at the 'House of Corkie'.



Instructions

Answer **all** questions in the spaces provided

In all questions where a numerical answer is required, an exact value must be given unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1 (6 marks)

Kreemond begins his long journey to the ‘House of Corkie’ and knows that he will arrive at the ‘House of Corkie’ on either Thursday morning or Friday morning. He also knows that he will be unlikely to have a meeting immediately, as Corkie is a very busy Oracle.

Corkie has three scheduled meeting times on Thursday: Afternoon, Twilight, Night.
He has two meeting times on Friday: Twilight and Night

If Kreemond arrives at the ‘House of Corkie’ on a Thursday morning, the probability of him having a meeting that afternoon is 0.3, the probability of meeting occurring during twilight is 0.6, otherwise the meeting will take place that night.

However, if he arrives at the ‘House of Corkie’ on a Friday morning the probability of him having a meeting at twilight is 0.4, otherwise the meeting will take place that night.

The probability of Kreemond arriving on Thursday morning is p .

a. Find, in terms of p the probability that:

i. The meeting does not occur at night.

2 marks

- ii. The meeting will occur in the afternoon, given that it does not occur at night.

Express your answer in the form $\frac{(k-1)p}{(k+1)p+k}$ where $k \in \mathbb{Z}$.

2 marks

- b. Find the value(s) of p such that the probability of the meeting not occurring at night is greater than 0.5.

2 marks



After his meeting with Corkie, Kreemond begins his journey back to JMSS.

Kreemond returns with a message from the oracle: the plague will end when the murderer of Mayus, the former king of JMSS, is caught and expelled; the murderer is within JMSS.

Schmidipus questions Kreemond about the murder of Mayus, who he believes was killed by a group of disgruntled students and educators on his way to get a coffee from the Campus Centre. Schmidipus promises to solve the mystery of Mayus' death, vowing to curse and drive out the murderer.

Schmidipus begins his investigation by questioning a large group of senior students.



Question 2 (12 marks)

Schmidipus, after conducting serious research has eleven main suspects for the murder. His suspects are 6 different year 12 students and 5 different year 11 students. Initially he plans to interview these students one at a time, so he decides to line them up in a single line in front of his office.

a. In how many different ways can these students be lined up so that:

- i.** The first 6 positions are occupied by year 12 students and the last 5 positions are occupied by year 11 students.

2 marks

- ii.** The first two and the last two positions are occupied by year 12 students.

2 marks

- iii.** All 5 year 11 students are **not** together in one group.

2 marks

After interviewing all the students individually, Schmidipus is particularly suspicious of a year 12 student named Raf and a year 11 student named Clara. Schmidipus decides that he might play some mind games and decides to interview the students in groups of four.

b. What is the probability that a random selected group of four will have:

i. No more than one year 11 student.

2 marks

ii. Either Raf or Clara, but not both in the group.

2 marks

iii. More year 11 students than year 12 students given that Raf is not in the group.

2 marks



Schmidipus begins to doubt that any of these students are responsible for Mayus' death and decides that he needs more help in solving this mystery. He sends for Tiresecic, the blind prophet, and asks him what he knows about the murder. Tiresecic, as always, responds cryptically. At first, he refuses to tell Schmidipus what he knows. Schmidipus curses and insults the Tiresecic, going so far as to accuse him of being a poor mathematician and a murderer. These taunts provoke Tiresecic into revealing that Schmidipus himself is the murderer.

Schmidipus naturally refuses to believe Tiresecic's accusation. He accuses Kreemond and Tiresecic of conspiring against him. He asks why Tiresecic did nothing when JMSS suffered under a plague once before. At that time, Xia-inx held the school captive and refused to leave until someone correctly answered her probability questions. Schmidipus brags that he alone was able to solve the probability questions posed by the monstrous Xia-inx.



Question 3 (12 marks)

- a.** For two events C and D , it is known that $\Pr(D) = q$ and $\Pr(C | D) = p^2$.
Find $\Pr(C' \cap D)$ in terms of p and q .

2 marks

- b.** A and B are independent events. $\Pr(A \cap B) = p$ and $\Pr(A) = 3\Pr(B)$.
Find $\Pr(A \cap B')$ in terms of p .

3 marks

c. A random variable T has the probability distribution given below.

t	1	2	3	4
$\Pr(T = t)$	n	$0.3 + n^2$	$0.2 - n^2$	$0.5 - n$

i. Find the possible value(s) of n . 2 marks

The mean of T is μ_T .

ii. State μ_T in terms of n . 1 mark

iii. Hence find the maximum possible value of μ_T . 2 marks

- iv. If $\Pr(T > 1 | T < 3) = 0.6$, find the value(s) of n . Give your answer(s) correct to three decimal places.

2 marks



Tiresecic defends his skills as a prophet, noting that Schmidipus' parents found him trustworthy. At this mention of his parents, Schmidipus, who grew up in the distant city of Bentleigh, asks how Tiresecic knew his parents. But Tiresecic, as always, answers enigmatically. Then, before leaving, Tiresecic puts forth one last riddle, saying that the murderer of Mayus will turn out to be both father and brother to his own children, and the son of his own wife.

Schmidipus explains to his wife Jokovska (also the widow of Mayus) that the prophet has accused him with Mayus' murder, and Jokovska replies that all prophecies are false. As proof, she notes that the oracle once told Mayus he would be murdered by his son, when in fact his son was cast out of JMSS as infant, and that Mayus was murdered by a group of educators. Her description of Mayus' murder, however, sounds familiar to Schmidipus, and he asks further questions. Jokovska tells him that Mayus was killed at the Campus Centre, just before Schmidipus arrived at JMSS. Schmidipus, stunned, tells his wife that he may be the one who murdered Mayus. He tells Jokovska that, long ago, when he was the prince of Bentleigh, he travelled to the oracle of Education, who told him he would murder his father and sleep with his mother. Hearing this, Schmidipus fled his home, never to return. It was then, on the journey that would take him to JMSS, that Schmidipus was confronted and harassed by a group of educators at the Campus Centre, whom he killed in self-defense. This skirmish occurred at the very Campus Centre where Mayus was killed.

To learn the truth, Oedipus sends a messenger to search for the only living witness to the murder, the ever-elusive McLeanus who was known to be a regular visitor to the Campus Centre.



Question 4 (9 marks)

A messenger is sent on a quest to find the ever-elusive McLeanus. The messenger knows that whilst making this journey he will meet with many travellers. He knows from experience that the probability of any traveller he meets being helpful in his quest is $\frac{1}{20}$ and this is independent of any other traveller being helpful.

- a.** In his first morning the messenger meets seven individual travellers.
Find the probability that from these seven travellers:

- i.** the first three travellers were helpful.

1 mark

- ii.** only the first three travellers were helpful. Express your answer in the form $\frac{a^m}{b^n}$, where $a, b, m, n \in \mathbb{Z}$.

2 marks

- iii.** Exactly three of the travellers were helpful. Give your answer correct to four decimal places.

2 marks

- iv.** At least one traveller was helpful. Give your answer correct to four decimal places. 2 marks

- b.** Find the minimum number of travellers that the messenger will have to meet to ensure that the probability of at least two travellers being helpful is greater than 0.98. 2 marks



Outside of JMSS, a messenger approaches Jokovska and tells her that he has come from Bentleigh to inform Schmidipus that his father, PollyFinch, is dead, and that the students of Bentleigh have asked Schmidipus to come and rule there in his place. Jokovska rejoices, convinced that PollyFinch's death from natural causes has disproved the prophecy that Schmidipus would murder his father. At Jokovska's summons, Schmidipus comes outside, hears the news, and rejoices with her. He now feels much more inclined to agree with the queen in deeming prophecies worthless and viewing chance as the principle governing the world. But while Schmidipus finds great comfort in the fact that one-half of the prophecy has been disproved, he still fears the other half—the half that claimed he would sleep with his mother.



Question 5 (10 marks)

Schmidipus spends a great deal of time worrying before he goes to sleep each night.

- a. The time, t , in hours, that Schmidipus spends worrying each night can be modelled by the random variable, T , with probability density function given by

$$T(t) = \begin{cases} \frac{1}{5} \sin(t) & 0 \leq t \leq \frac{\pi}{2} \\ mt + \frac{1}{5} & \frac{\pi}{2} < t \leq 2 \end{cases}$$

Calculate the value of m .

2 marks

After seeking some advice for his wellness, Schmidipus' time spent worrying each night changes. The time, t , in hours, that Schmidipus spends worrying each night can now be modelled by the new random variable, M , with probability density function given by

$$M(t) = \begin{cases} \frac{2}{5} \sin\left(\frac{\pi t}{2}\right), & 0 \leq t \leq 1 \\ \left(\frac{8\pi - 8}{15\pi}\right)t + \frac{1}{5}, & 1 < t < 2 \end{cases}$$

b. Find, correct to four decimal places:

- i.** The probability that Schmidipus spends less than 90 minutes worrying on any particular night.

1 mark

- ii.** The probability that Schmidipus spends less than 90 minutes worrying on any particular night, given that he spends more than 30 minutes worrying.

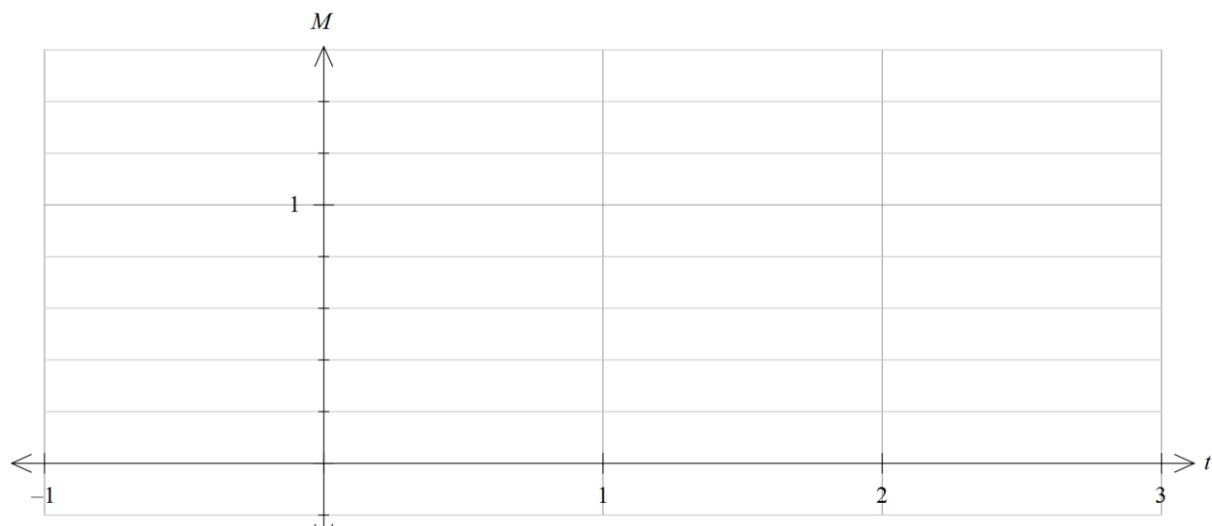
2 marks

- iii. The value of t , correct to the nearest minute, such that $\Pr(M < t) = 2\Pr(M > t)$.

2 marks

- c. Sketch the graph of $M(t)$ on the set of axes below. Label all endpoints with their coordinates. State any irrational values correct to two decimal places.

3 marks



The messenger remarks that Schmidipus need not worry, because PollyFinch and his wife, MeropHale, are not Schmidipus' biological parents. The messenger, a tutor by profession, knows firsthand that Schmidipus came to Bentleigh as an orphan. He explains that one day long ago, he was tutoring his students when another tutor approached him carrying a baby. The tutor took the baby to the royal family of Bentleigh, and they raised him as their own. That baby was Schmidipus. Schmidipus asks who the other tutor was, and the messenger answers that he was a past student of Mayus.

All of this talk of tutors causes Jokvska to run back into her office at JMSS in despair.



Question 6 (13 marks)

Jokovska has a fascination with the hourly rate of tutoring that students at JMSS pay.

She knows that the hourly rate, H , in dollars, that Year 11 students pay is a random variable that follows a normal distribution with mean μ_H and standard deviation σ_H .

She also knows that $\Pr(H < 70) = 0.2477$ and $\Pr(H > 110) = 0.1279$.

- a. Find, correct to the nearest cent, the values of μ_H and σ_H .

2 marks

Jokovska also knows that the hourly rate, K , in dollars, that Year 12 students pay also follows a normal distribution with mean $\mu_K = 90$ and standard deviation $\sigma_K = 25$.

- b. Use the **68-95-99.7 percent rule** and **appropriate diagrams** to find the following:

- i. The approximate percentage, correct to one decimal place, of Year 12 students that pay an hourly rate between \$40 and \$90.

2 marks

- ii. The approximate value of a , correct to three decimal places, such that $\Pr(K > a)$ is approximately 0.975.

2 marks

- c. Find correct to four decimal places, the probability that:

- i. A Year 12 student pays an hourly rate of more than \$100.

1 mark

- ii. A Year 12 student pays an hourly rate of more than \$100, given that they pay less than \$110.

2 marks

- iii. Two siblings, one in Year 11 and one in Year 12 both pay more than \$110 per hour each for their tutoring.

2 marks

- iv. From a group of five Year 12 students who all pay for tutoring, at least three of them pay more than \$100 per hour for tutoring.

2 marks



Schmidipus then approaches the tutor and interrogates him, asking who gave him the baby. The tutor refuses to disclose anything, and Schmidipus threatens to restrict his access to JMSS's IP. Finally, he answers that the child came from the house of Mayus. Questioned further, he answers that the baby was in fact the child of Mayus himself, and that it was Jokovska who gave him the infant, ordering him to kill it, as it had been prophesied that the child would kill his parents. But the tutor pitied the child and decided that the prophecy could be avoided just as well if the child were to grow up in a foreign city, far from his true parents. The tutor therefore passed the boy on to the tutors in Bentleigh.



Question 7 (3 marks)

The number of JMSS prophecies that have come to fruition in the time of Schmidipus can be considered a binomial random variable with a mean of 40 and a variance of $\frac{20}{3}$.

Find, correct to four decimal places, the probability that at least 40 prophecies come to fruition.



Realizing who he is and who his parents are, Schmidipus screams that he sees the truth and flees back into his office. The tutor and the messenger slowly leave JMSS.



Question 8 (7 marks)

The tutor explains to Jcovska that the amount of time, t , that he spends tutoring a particular JMSS student each session can be modelled by the random variable, R , with probability density function given by

$$R(t) = \begin{cases} \frac{3e^{t-2}}{10}, & 1 \leq t \leq k \\ 0 & \text{otherwise} \end{cases}.$$

- a.** Find the value of k . Give your answer in the form $\log_e \left(\frac{ne + (n^2 + 1)e^2}{n} \right)$ where $n \in \mathbb{Z}$. 2 marks

The tutor decides to test the mathematical acumen of the messenger by asking the following questions.

- b.** For a particular tutor, the following is known regarding the length of time measured in minutes of a tutoring session:

For year 11 students the time is a random variable, X , that follows a normal distribution with a mean of 60 minutes and a standard deviation of 10 minutes.

For year 12 students the time is a random variable, Y , that follows a normal distribution with a mean of 70 minutes and a standard deviation of 12 minutes.

Find, correct to four decimal places, the probability that the length of time of a tutoring Session for a randomly selected year 12 student is longer than 75% of the length of times of a tutoring session for year 11 students.

2 marks

- c.** Let Q be a random variable that follows a normal distribution with a mean 10 and standard deviation 3. Let Z be the standard normal random variable.

Find the value of a such that $\Pr(Z < 2a) = \Pr\left(Q > \frac{a-1}{10}\right)$.

3 marks



Schmidipus now emerges from JMSS, begging to be exiled. He asks Kreemond to send him away from JMSS.

Kreemond, covetous of royal power, is all too happy to oblige. And so, despite his precautions, the prophecy that Schmidipus dreaded has actually come true. Realizing that he has killed his father and married his mother, Schmidipus is agonized by his fate.

